An Asymptotic Correction to Luce's Law for Competing Cox Processes

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Abstract

Consider N channels that fire at random times, with rates driven by a hidden state they share, so that conditionally on that state each is Poisson and the whole is a doubly stochastic process with N marks. Only the first arrival is recorded. When the driver mixes faster than the channels fire, the probability that channel i arrives first converges to Luce's law in the stationary mean rates. We give the first correction. It is governed by the time-integrated covariance K of the rates, a dynamical rather than a static quantity, and it is what breaks independence of irrelevant alternatives. One pair of objects then predicts every blocked-set counterfactual by restricting sums, a driver with r modes gives K of rank r, and fluctuations common to all channels cancel exactly. We then ask what observation reveals about K, and it is identified only in part, modulo an explicit $(N+1)$-dimensional family. Recording the second arrival as well as the first reaches every identified combination, matching what blocking channels and rerunning would give, while first-arrival data alone reaches $N-1$ of them.

1 Introduction

Consider N channels that fire at random times, whose rates are not constants but are driven by a hidden state all of them share. Conditional on the path of that state each process is Poisson, so this is a doubly stochastic, or Cox, process in the sense of Cox (1955), with N marks and a common driver. Only the first firing is recorded. Chemistry supplies the picture, in which a reaction escapes through one of several channels while the surrounding configuration rearranges underneath, and the same structure appears whenever a kinetic Monte Carlo (KMC) step selects a move (Bortz et al., 1975; Gillespie, 1977) or a diffusing particle finds one of several windows.

The shared driver is the interesting part and the unobservable one. It makes the channels' rates fluctuate together, which is what breaks independence of irrelevant alternatives (IIA). Under IIA a removed stream's share is redistributed in proportion to the survivors', and where that has been measured it fails, in diversion ratios estimated from demand data (Conlon and Mortimer, 2021), in choice experiments with shared features (Tversky, 1972), and in the specification test built on exactly this restriction (Hausman and McFadden, 1984). The driver is also exactly what an experimenter cannot see. So the question is how much of it can be recovered from the arrivals themselves.

*Code and machine verification for every claim in this paper: `https://github.com/microprediction/kinetics` (experiments 7 and 9). Web companion: `https://kinetics.microprediction.org`.

Our main result is that the answer is a great deal, and from surprisingly little data. Recording who won and who came second, from the full set alone, recovers everything about the driver that the choice law can ever reveal. It matches what one would learn from running every possible experiment in which some channels are blocked and the process is rerun. Winner-only data, by contrast, recovers almost nothing, $N-1$ numbers where the full answer has N^2-N-1. Finishing orders are usually recorded already and interventions are usually impossible, so this is the difference between a feasible measurement and an infeasible one.

Making that precise requires knowing how the driver enters the choice law, and that is a homogenization question. When the driver mixes faster than the channels fire, the winning probabilities converge to softmax in the stationary mean intensities. The first correction is governed by

$$K_{jk} = \int_0^\infty \mathrm{Cov}_\pi(\lambda_j(Y_0), \lambda_k(Y_t))\, dt, \tag{1}$$

the time-integrated covariance of the firing rates. It is a dynamical quantity. Two channels whose rates move together at any instant contribute nothing unless that agreement persists over the time the driver takes to forget itself.

K is therefore the object to be identified, and it is identified only in part. Choice probabilities determine K modulo an explicit family of dimension $N+1$, leaving N^2-N-1 combinations knowable and the rest permanently invisible. The invisible directions are not a defect of any experiment. They are the ones that, like a rate shift applied to every channel at once, move no branching ratio and so change nothing an observer of winners could ever see.

The bridge from second place to K is an identity that costs nothing. Deleting a channel changes the outcome only in those realizations the deleted stream arrived first, and in those the runner-up inherits, so $q_j^{(-i)} = p_j + p_i M_{ij}$ exactly, at every degree of timescale separation. Second-place frequencies are thus deletion experiments that have already been run.

Three further properties make the correction usable. One pair $(\bar{\lambda}, K)$ serves every availability set, so the whole deletion ensemble follows from a single calculation. Fluctuations shared by all channels cancel exactly, which is the proportional-hazards normal form and is classical (Elandt-Johnson, 1976; Marley and Colonius, 1992). And a driver with r modes gives K of rank r.

Section 2 sets up the arrival process and its killed resolvent. Section 3 proves the homogenization expansion and places it against the motional-narrowing literature, where the diagonal of K has been known for seventy years. Section 4 collects the structural properties, Section 5 the second-place identity, and Section 6 the identification and design results. Sections 7 and 8 verify everything, first on finite chains where every quantity is an exact linear solve, then on reflected Brownian motion in a disk with boundary reaction windows.

2 Competing arrivals under a shared driver

Let Y_t be an ergodic Markov process on a state space with generator L/ε and invariant law π. The parameter ε is the ratio of the environment's mixing time to the timescale on which channels fire. Small ε means fast mixing.

Each channel $i \in \{1, \ldots, N\}$ fires at intensity $\lambda_i(Y_t) \geq 0$, conditionally Poisson given the path of Y. The first channel to fire wins. Conditionally on the driver the arrivals are Poisson, so the whole is a doubly stochastic Poisson process (Cox, 1955) with N marks, observed until its first event. When Y is a finite chain it is the marked Markov-modulated Poisson process of the queueing literature (Fischer and Meier-Hellstern, 1993), and the λ_i are cause-specific hazards in the competing-risks sense. Note this is an arrival process and not a Thurstonian race. Nothing here is an argmin of latent performances. For an available set $A \subseteq \{1,\dots,N\}$ write $\Lambda_A(y) = \sum_{j\in A} \lambda_j(y)$ for the total hazard, and write $\bar\lambda_i = \int \lambda_i \, d\pi$ for the stationary mean intensity.

Let $u_i^A(y)$ be the probability that channel i wins, started from $Y_0 = y$. Conditioning on the first instant and using the conditional Poisson structure gives

$$\left(\frac{L}{\varepsilon} - \Lambda_A\right) u_i^A = -\lambda_i, \qquad u_i^A = \left(\Lambda_A - \frac{L}{\varepsilon}\right)^{-1} \lambda_i. \tag{2}$$

This is a killed resolvent. The generator propagates the environment and the multiplication operator Λ_A kills paths at the total firing rate. On a finite state space it is a linear system, which is what makes the verification in Section 7 exact.

The observable of interest is the win probability from a stationary start,

$$p_i^A = \int u_i^A \, d\pi. \tag{3}$$

Summing (2) over $i \in A$ gives $\sum_{i\in A} u_i^A = 1$, so the p_i^A are a probability vector on A for every ε.

3 Homogenization and the Green–Kubo correction

The expansion below is the averaging principle for a fast ergodic driver (Khas'minskii, 1966) in the operator form of Kurtz (1973), and its solvability conditions are the usual cell problem (Pavliotis and Stuart, 2008, Ch. 9). Write $\tilde\lambda_i = \lambda_i - \bar\lambda_i$ for the fluctuation of the ith intensity, and Π for the projection $\Pi f = (\int f d\pi)\mathbf{1}$ onto constants. The deviation operator

$$D = (\Pi - L)^{-1}(I - \Pi) \tag{4}$$

inverts L on the mean-zero subspace, and $Dg = \int_0^\infty e^{tL} g \, dt$ for mean-zero g. Note the sign it carries. Differentiating that integral gives $LDg = -g$, so D is minus the pseudo-inverse of L, and the sign propagates into the proof below. The Green–Kubo matrix is

$$K_{jk} = \int_0^\infty \mathrm{Cov}_\pi(\lambda_j(Y_0), \lambda_k(Y_t)) \, dt = \int \tilde\lambda_j \, (D\tilde\lambda_k) \, d\pi. \tag{5}$$

The second expression is what one computes. It replaces a time integral by a single linear solve. Note that K need not be symmetric. It is symmetric when the environment is reversible, and the verification in Section 7 deliberately uses a non-reversible chain so that the index placement in the theorem is actually tested.

The diagonal of K is not new. For a single channel with a fluctuating rate the integral $\int_0^\infty \mathrm{Cov}_\pi(\lambda(Y_0), \lambda(Y_t)) \, dt$ is the classical second-order cumulant correction to an effective decay rate, known since Anderson (1954) and Kubo (1954) as motional narrowing and developed as a general expansion by van Kampen (1974). The name Green–Kubo is used here in the sense of Green (1954) and Kubo (1957), an integrated dynamical correlation standing in for a transport coefficient.

What that scalar theory cannot see is the object this paper is about. A rate correction common to all channels cancels from a ratio, so it leaves the choice law untouched. The branching ratio is moved instead by the off-diagonal and generally non-symmetric entries K_{ji}, and those are what Theorem 1 puts to work.

Theorem 1 (Homogenized choice law and its first correction)**.** *Let* $\bar\Lambda_A = \sum_{j\in A} \bar\lambda_j$ *and suppose* $\bar\Lambda_A > 0$. *Then as* $\varepsilon \to 0$,

$$p_i^A = \frac{\bar\lambda_i}{\bar\Lambda_A} - \frac{\varepsilon}{\bar\Lambda_A}\left[\sum_{j\in A} K_{ji} - \frac{\bar\lambda_i}{\bar\Lambda_A}\sum_{j,k\in A} K_{jk}\right] + O(\varepsilon^2). \tag{6}$$

In particular $p_i^A \to \bar\lambda_i/\bar\Lambda_A$, *the softmax law in the log mean intensities* $\theta_i = \log\bar\lambda_i$.

The limit is worth naming carefully, because the same law arrives in this literature by three different routes. It is Luce's choice model (Luce, 1959), the conditional logit of McFadden (1974), and the softmax output layer of Bridle (1990). In the random-utility tradition it follows from Gumbel noise, and Yellott (1977) showed that representation is the only one consistent with Luce's axiom under independent discriminal processes. Here it arrives from neither an axiom nor a noise family but from fast mixing, and for constant rates it is already in Cox (1959). A reader should take $\theta_i = \log\bar\lambda_i$ as a relabelling rather than a derived exponential family.

Proof. Multiply (2) by ε to get $(L - \varepsilon\Lambda_A)u_i^A = -\varepsilon\lambda_i$, expand $u_i^A = u_i^{(0)} + \varepsilon u_i^{(1)} + \varepsilon^2 u_i^{(2)} + O(\varepsilon^3)$, and collect powers of ε.

At order ε^0, $Lu_i^{(0)} = 0$, so $u_i^{(0)}$ is a constant c_i by ergodicity.

At order ε^1 the equation reads $Lu_i^{(1)} = \Lambda_A c_i - \lambda_i$. A solution exists only if the right side is orthogonal to π, which forces $\bar\Lambda_A c_i = \bar\lambda_i$ and gives the leading term. The right side is then $c_i\tilde\Lambda_A - \tilde\lambda_i$ with $\tilde\Lambda_A = \sum_{j\in A}\tilde\lambda_j$, and inverting with $LD = -I$ on mean-zero functions gives

$$u_i^{(1)} = -D\Big(c_i\tilde\Lambda_A - \tilde\lambda_i\Big) + m_i,$$

where the constant $m_i = \int u_i^{(1)} d\pi$ is fixed at the next order.

At order ε^2 the equation is $Lu_i^{(2)} = \Lambda_A u_i^{(1)}$, whose solvability condition is $\int \Lambda_A u_i^{(1)}\, d\pi = 0$. The constant part of Λ_A contributes $\bar\Lambda_A m_i$, the constant m_i integrates against $\tilde\Lambda_A$ to zero, and what remains is

$$0 = \bar\Lambda_A m_i - \int \tilde\Lambda_A\, D\Big(c_i\tilde\Lambda_A - \tilde\lambda_i\Big)\, d\pi.$$

Using (5), $\int \tilde\Lambda_A D\tilde\lambda_i\, d\pi = \sum_{j\in A} K_{ji}$ and $\int \tilde\Lambda_A D\tilde\Lambda_A\, d\pi = \sum_{j,k\in A} K_{jk}$, so

$$m_i = -\frac{1}{\bar\Lambda_A}\left[\sum_{j\in A} K_{ji} - \frac{\bar\lambda_i}{\bar\Lambda_A}\sum_{j,k\in A} K_{jk}\right].$$

Finally $p_i^A = c_i + \varepsilon m_i + O(\varepsilon^2)$ by (3), which is (6). □

Proposition 1 (General start)**.** *Let the environment start from an arbitrary law* μ_0 *rather than from* π, *and write* $p_i^A(\mu_0) = \int u_i^A\, d\mu_0$. *Then*

$$p_i^A(\mu_0) = \frac{\bar\lambda_i}{\bar\Lambda_A} + \varepsilon\left[m_i - \int D\left(\frac{\bar\lambda_i}{\bar\Lambda_A}\tilde\Lambda_A - \tilde\lambda_i\right) d\mu_0\right] + O(\varepsilon^2), \tag{7}$$

with m_i *the constant appearing in the proof of Theorem 1. The leading softmax term does not depend on the start. The extra term vanishes when* $\mu_0 = \pi$, *because* D *maps into mean-zero functions, which recovers* (6).

Proof. The expansion in the proof of Theorem 1 is pointwise in y, and integrating $u_i^{(1)} = -D(c_i\tilde{\Lambda}_A - \tilde{\lambda}_i) + m_i$ against μ_0 instead of π leaves the displayed extra term. □

This is worth stating because it is easy to apply (6) to a system observed away from equilibrium and silently lose the correction. Experiment 40 measures the loss. From a point-mass start the stationary formula (6) is accurate only to order 0.968, while (7) restores order 1.963. Under the invariant law the extra term is 7.3×10^{-17} and the two forms agree.

Remark 1. The two terms in the bracket have distinct roles. The first, $\sum_{j\in A} K_{ji}$, measures how channel i's rate correlates dynamically with the total hazard it competes against. The second subtracts the part of that effect which is common to all channels, and is what keeps $\sum_{i\in A} p_i^A = 1$ at each order.

4 Three structural properties

One computation answers every counterfactual. The pair $(\bar{\lambda}, K)$ does not depend on A. Every blocked-set counterfactual is obtained from (6) by restricting the sums to the surviving set. For a system with N channels this reduces 2^N separate resolvent solves to one deviation solve and $O(N^2)$ stored numbers, at the cost of the $O(\varepsilon^2)$ error the expansion carries.

Common-mode fluctuations cancel. Suppose $\lambda_i(y) = a_i c(y)$, so that every channel is modulated by the same environmental factor. Then $\tilde{\lambda}_i = a_i\tilde{c}$ and $K_{jk} = a_j a_k \gamma$ with $\gamma = \int \tilde{c}\,(D\tilde{c})\,d\pi$. Substituting into the bracket in (6) gives $a_i\gamma \sum_{j\in A} a_j - (a_i/\sum_{j\in A} a_j)\gamma(\sum_{j\in A} a_j)^2 = 0$. The correction vanishes identically.

Proposition 2 (Proportional hazards give Luce exactly)**.** *Suppose $\lambda_i(y) = a_i c(y)$ with $a_i > 0$ and $c \geq 0$ not identically zero. Then for every y, every ε, and every A,*

$$u_i^A(y) = \frac{a_i}{\sum_{j\in A} a_j},$$

so the win probability is Luce's law from any start, not merely from π and not merely to the order computed above.

This is a known result in a new setting rather than a new one, and the prior statements are worth being precise about. Elandt-Johnson (1976) proved that under proportional hazard rates the failure time and the cause of failure are independent, and did so without assuming the latent failure times are independent. Kochar and Proschan (1991) give an equivalent statement. Marley and Colonius (1992) prove the choice-theoretic version as an equivalence, that proportional cause-specific hazards hold if and only if the chosen option and the time of choice are independent, with the constants of proportionality forced to equal the choice probabilities. They also record the explicit representation $\Pr[t(x) > t] = \exp(-u(x)\Psi(t))$, which yields $u(x)/\sum_y u(y)$ for an arbitrary increasing Ψ.

What the present setting adds is only that the common factor may be a random functional of the hidden environment rather than a deterministic function of time, and the proof below is correspondingly short.

Proof. Let v_i be the constant function $a_i/\sum_{j\in A} a_j$. Constants are annihilated by L, and $\Lambda_A v_i = c(y)\big(\sum_{j\in A} a_j\big) \cdot a_i/\sum_{j\in A} a_j = a_i c(y) = \lambda_i$, so $(\Lambda_A - L/\varepsilon)v_i = \lambda_i$. Since the killed resolvent is invertible, v_i is the unique solution of (2), which is u_i^A. □

A common multiplicative modulation is a shared frailty in the survival sense (Vaupel et al., 1979), and the invariance it produces is the proportional-hazards structure of Cox (1972). The proof needs no positivity of c, no time change, and no condition on ε. What it says is that a common modulation factors out of the competition entirely, which is the proportional-hazards normal form. Experiment 40 confirms it where the hypotheses are stressed. With c vanishing on four of six states, with c ranging over nine orders of magnitude, and at $\varepsilon = 100$, the departure from Luce stays below 5.5×10^{-14} at every state.

This is a stronger statement than the vanishing of the correction, and it is practically useful in the opposite direction. It says a measured departure from softmax cannot be produced by any amount of shared environmental fluctuation. Whatever deviation is observed is evidence of differential structure among the channels. It is worth setting this against the classical negative results. The latent joint law of competing risks is not identified from what one observes (Tsiatis, 1975), and identification must be bought with extra structure such as regressor variation (Heckman and Honoré, 1989). The statement here runs the other way. One large class of environmental dependence, the perfectly shared one, leaves no trace whatever in the choice law, so any trace that does appear rules that class out.

Low-rank environments give low-rank corrections. Suppose the intensity deviations are driven by r environmental modes, $\lambda_i - \bar{\lambda}_i = b_i^\top z(y)$ with z taking values in $\mathbb{R}^r$ and $\int z \, d\pi = 0$. Then $K = B\Gamma B^\top$ with B the $N \times r$ matrix of loadings and $\Gamma_{ab} = \int z_a(Dz_b) d\pi$, so K has rank at most r. There is a second cap. Deviations live in the mean-zero subspace of the environment, so $\operatorname{rank}(K) \le \min(N, m-1)$ for an environment with m states, whatever the loadings are. Experiment 40 attains both bounds, giving rank exactly r for $r = 1, \dots, 5$ on a six-state environment and rank $5 = \min(10, 5)$ for generic intensities on ten channels. The compressed correlated competition that appears elsewhere in this program therefore has a physical provenance rather than being a modelling convenience. The same low-dimensional error-components structure is imposed by hand in mixed logit (McFadden and Train, 2000), where it is a specification choice. Here it is inherited from the environment.

5 The correction predicts the runner-up

The correction has been described so far as a statement about win probabilities, which are what a blocked-set experiment measures. Those experiments are expensive, since each availability set requires its own intervention. There is a cheaper observable that carries the same information.

Let $M_{ij} = \mathbb{P}(\text{runner-up} = j \mid \text{winner} = i)$ be the runner-up kernel, the probability that j finishes second given that i finishes first. Removing channel i changes the outcome only in those realizations where i won, and in those the runner-up inherits the win. This gives an identity that holds for every ε and involves no approximation,

$$q_j^{(-i)} = p_j + p_i M_{ij}, \tag{8}$$

where $q^{(-i)}$ is the deletion counterfactual. The companion work shows M is exactly the object that identifies single deletions, and that winner-only data constrains it not at all.

Applying Theorem 1 to the full set and to $A \setminus \{i\}$ therefore predicts M itself. At leading order the prediction is

$$M_{ij} = \frac{\bar{\lambda}_j}{\bar{\Lambda}_A - \bar{\lambda}_i} + O(\varepsilon), \tag{9}$$

which says the second arrival is drawn from the competition among the remaining channels, the IIA prediction. Substituting the corrected shares into (8) gains an order.

Experiment 39 verifies this on the same chains. The identity (8) holds to 1.8×10^{-15}, the exactly computed kernel is row-stochastic to 8.0×10^{-15}, which checks the two-stage solve, and the measured convergence orders are 0.967 for the Luce prediction (9) and 1.967 with the Green–Kubo correction. At $\varepsilon = 0.005$ the maximum error over off-diagonal entries falls from 3.0×10^{-4} to 2.1×10^{-6}.

This matters for testing the theory. The second arrival is observed wherever an order of arrivals is recorded, so the correction can be checked against ordinary ranked data rather than against interventions. It also runs the other way. Where the environment is unknown, a measured M constrains the Green–Kubo matrix, which is the beginning of estimating K from data rather than computing it from a known generator.

6 What data reveals about K

Everything so far computes K from a known generator. A practitioner rarely has the generator. They have observed arrivals. This section asks what K can be recovered from outcomes alone, and the answer has a clean form.

The correction in (6) is linear in K, so stacking it over every availability set gives a linear map from K to observable deviations from softmax. Its null space is exactly what no amount of choice data can reveal.

Proposition 3 (Identifiability of K). *Choice probabilities on all availability sets determine K exactly modulo the $(N+1)$-dimensional family*

$$K \longmapsto K + d\,\bar{\lambda}^{\top} + t\,\mathrm{diag}(\bar{\lambda}), \qquad d \in \mathbb{R}^N,\ t \in \mathbb{R}, \tag{10}$$

and no smaller family. Exactly $N^2 - N - 1$ combinations of the entries of K are identified.

Proof. Write $B(A,i) = \sum_{j\in A} K_{ji} - c_i \sum_{j,k\in A} K_{jk}$ for the bracket in (6), with $c_i = \bar{\lambda}_i/\bar{\Lambda}_A$. Under $K \mapsto K + d\bar{\lambda}^{\top}$ the first sum gains $\bar{\lambda}_i \sum_{j\in A} d_j$ and the second gains $\bar{\Lambda}_A \sum_{j\in A} d_j$, so the change in B is $\bar{\lambda}_i \sum_{j\in A} d_j - c_i\bar{\Lambda}_A \sum_{j\in A} d_j = 0$. Under $K \mapsto K + t\,\mathrm{diag}(\bar{\lambda})$ the first sum gains $t\bar{\lambda}_i$, since only the $j = i$ term contributes, and the second gains $t\bar{\Lambda}_A$, so the change is $t\bar{\lambda}_i - c_i t\bar{\Lambda}_A = 0$. The two families intersect trivially and span $N+1$ dimensions. That the null space is no larger is a rank computation, carried out in experiment 41 for $N = 3, \ldots, 6$, where the identified dimension equals $N^2 - N - 1$ in every case. $\square$

The two invisible directions have meaning. Adding $d\bar{\lambda}^{\top}$ changes how each channel's fluctuations correlate with the environment in the direction of the mean intensities, and adding $t\,\mathrm{diag}(\bar{\lambda})$ rescales each channel's own autocorrelation in proportion to its mean rate. Neither moves any branching ratio, for the same reason a common rate shift does not.

Which experiments reach the identified part. The design matters far more than the sample size, and the useful comparison is between three ways of watching the same system. Winner-only data on the full set is nearly useless for this purpose. It supplies $N-1$ independent numbers, so it recovers $N-1$ of the $N^2 - N - 1$ identified directions and no more. Running every blocked-subset experiment recovers all of them, which is the content of Proposition 3.

Between those sits the observation that makes the theory testable. Recording the runner-up alongside the winner, on the full set alone, recovers *everything* that the complete set of blocked-subset interventions would. Experiment 41 confirms this for $N = 4, 5, 6$, where the ranked design and the exhaustive intervention design have identical rank, $N^2 - N - 1$, while the winner-only design has rank $N - 1$. Interventions are often impossible, and a finishing order is usually free.

What it costs. Identification is not estimation. Fitting K by least squares on the identified subspace, from shares estimated on pairs and triples and then scored on held-out larger subsets, the estimator beats the uncorrected softmax law only once roughly 10^6 observed arrivals per availability set are observed. The reason is a floor rather than a defect of the estimator. The correction is $O(\varepsilon)$ while multinomial noise on n observations is $O(n^{-1/2})$, so the data requirement scales as ε^{-2}. At the $\varepsilon = 0.05$ of experiment 41 the correction is 1.2×10^{-3} and the crossover duly appears between 10^5 and 10^6 observations.

This is worth stating plainly because it bounds the practical claim. Where the generator is known, the correction is cheap and accurate. Where only outcomes are observed, recovering it demands either very many observations or the richer observable of Section 5.

7 Verification on finite chains

Everything above is structural. This section and the next test it numerically. On a finite state space every object is an exact linear solve, so the expansion can be checked without simulation error. Experiment 7 runs a six-state chain with ten channels.

Setup. The generator has independent unit-mean exponential off-diagonal rates, giving a non-reversible chain with spectral gap 2.63. Channel intensities are lognormal with log-scale 0.8, giving $\bar{\Lambda}_A = 11.4$ on the full set. The seed is 5 and the sweep covers thirteen values of ε from 10^{-3} to 1. The dimensionless small parameter is $\varepsilon\bar{\Lambda}_A/\text{gap} \approx 4.3\,\varepsilon$, so the fitted window ends near 0.08 in those units and $\varepsilon = 0.3$ corresponds to 1.3.

The measured convergence order of the leading term is 0.985, against the theoretical 1. After adding the Green–Kubo correction the measured order is 1.985, against the theoretical 2. Both are fitted over the smallest six of thirteen values of ε, spanning 10^{-3} to 1.8×10^{-2}, since the expansion is asymptotic and the largest ε carry the $O(\varepsilon^2)$ residual. The gain of exactly one order is the signature the expansion predicts.

One instance is an anecdote, so experiment 40 repeats the measurement over twenty environments drawn with four to nine hidden states, three to eight channels, and intensity dispersions from 0.3 to 1.2. The leading order has median 0.960 and range $[0.919, 0.982]$, the corrected order has median 1.958 and range $[1.888, 1.979]$, and the gain exceeds 0.948 in every single trial.

Figure 1 shows both. The subset-uniformity claim is tested by fixing one $(\bar{\lambda}, K)$ pair and predicting every availability set. Experiment 40 sweeps all 1013 subsets of size at least two at $\varepsilon = 0.05$, so the maxima here are true maxima rather than samples. The worst error over all subsets falls from 3.1×10^{-3} under softmax to 4.4×10^{-4} with the correction, with no per-subset computation.

The order is uniform in a strong sense. Fitting the convergence order separately on each of the 1013 subsets puts every corrected order in $[1.92, 2.00]$. Small subsets are better rather than worse, because the effective parameter is $\varepsilon\bar{\Lambda}_A/\text{gap}$ and $\bar{\Lambda}_A$ shrinks with the subset. Exactly one

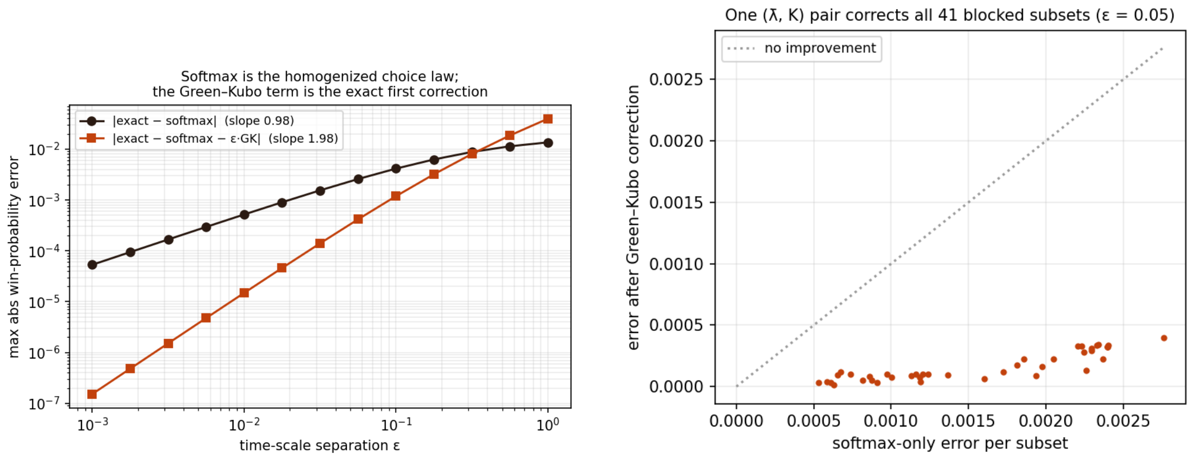

Figure 1: Finite-chain verification (experiment 7). Left, error against ε for the leading softmax law and for the Green–Kubo corrected law, with measured orders 0.985 and 1.985. Right, the same $(\bar{\lambda}, K)$ pair applied to 41 random availability sets at $\varepsilon = 0.05$, with no per-subset computation.

subset has the correction slightly worse than softmax, a two-element near-tie where both errors are already two orders of magnitude below the maximum.

The word Green–Kubo is doing work in (5), and experiment 40 tests whether it needs to. Replacing K by the equal-time covariance $\int \tilde{\lambda}_j \tilde{\lambda}_k \, d\pi$ scaled by a single timescale, with that timescale fitted to give the ablation its best case, leaves the measured order at 1.09 rather than 1.96, and the error at the smallest ε is 8.8 times worse. Static correlation between two channels does not determine the correction. The time integral does.

Common-mode cancellation is confirmed at 2.6×10^{-16} for the Green–Kubo coefficient, and the exact win probabilities agree with Luce to 1.1×10^{-16} at $\varepsilon = 0.3$. The regression suite carries the same check at $\varepsilon = 3.0$, where the environment is ten times slower than the channels rather than faster and the expansion has no claim at all, and Luce still holds to 10^{-12}. That is the content of Proposition 2. Rank-two environmental loadings give K with singular values $[0.022, 0.002, 0, 0]$, so the rank is exactly two.

A Gillespie simulation of the same system, event-driven and including the hidden state, agrees with the resolvent solution to 1.9×10^{-3} across 40,000 trajectories at $\varepsilon = 0.3$, against a binomial sampling noise of 2.1×10^{-3}.

The chain is not reversible, so K is asymmetric. On this instance $\max_{jk} |K_{jk} - K_{kj}| = 0.139$ against $\max_{jk} |K_{jk}| = 0.603$, a relative asymmetry of 23% recorded by experiment 40. This is what distinguishes K_{ji} from K_{ij} in (6), and experiment 40 confirms the index order is not a convention. Transposing K in the correction degrades the measured order from 1.94 to 1.58, and K agrees with brute-force time integration of the covariance to 4.1×10^{-6}.

8 Narrow escape

Finite chains verify the algebra. To test the expansion as a statement about physics we take the hidden state to be reflected Brownian motion in the unit disk and the channels to be reaction windows on the boundary, bands in which channel i fires at rate κ. This is the narrow escape geometry (Schuss et al., 2007) with partially reactive patches, and the leading band-area law is the well-mixed limit of Berg and Purcell (1977). The first window to fire wins. Here κ plays the role of ε, since a slow reaction rate means the particle mixes many times before firing.

The correspondence with Section 2 is worth spelling out, since the driver here is the physical system itself rather than an external environment. The hidden state Y_t is the particle's position in the disk, and its generator is the reflecting Laplacian. The channels are the windows, and window i fires at rate $\lambda_i(y) = \kappa\, g_i(y)$, where g_i is the fraction of the neighbourhood of y lying inside window i's band, so λ_i is supported near the boundary and vanishes in the interior. Two consequences follow. The mean rates $\bar{\lambda}_i$ are proportional to band areas, which is why the leading law is band-area softmax. And K is built from how long the particle's excursions keep it near one window rather than another, which is a purely geometric quantity.

The small parameter is κ rather than ε. Slow reaction means the particle explores the whole disk many times before anything fires, so small κ is the fast-mixing limit, and blocking a channel means switching a window off.

Setup. Sixteen boundary windows on the unit circle, six clustered within about 50 degrees and the rest scattered, drawn with seed 42. Angular half-widths are lognormal with median 0.030, ranging over $[0.019, 0.055]$ on this draw, and three pairs overlap. The reaction bands extend 0.12 inward in radius. The polar grid is 40×88, giving 3520 cells. The resolvent sweep covers $\kappa \in \{2, 5, 10, 20, 40\}$, and the Brownian simulation runs at $\kappa \in \{5, 20\}$ with time step 2×10^{-4} and 50,000 trajectories.

Experiment 9 separates three layers so that no error source can hide in another. Layer A is a real Brownian simulation with independent per-step firing. Layer B is an exact killed-resolvent solve on a finite-volume Neumann Laplacian over a polar grid, with the uniform invariant law, reversibility, and conservation checked numerically. Layer C is the theory, band-area softmax plus the Green–Kubo term from one deviation solve, using the same code that passed the chain tests.

Two homogenizations should be kept apart here. Replacing a patchy boundary by an effective uniform reactivity is boundary homogenization (Berezhkovskii et al., 2004; Lawley and Keener, 2015), and expansions in a small geometric parameter give splitting probabilities among small targets (Bressloff, 2020). The small parameter in this paper is temporal rather than geometric, so the two expansions are orthogonal and compose.

Comparing layers B and C tests the theorem on a continuous-physics generator. The softmax error decays with measured order 0.93 and the corrected error with order 1.93, again a clean gain of one order. Both orders are fitted on the three smallest values of κ, for the same reason as before.

Table 1 gives the full sweep, including where the correction stops paying. The improvement factor falls monotonically from $31\times$ at $\kappa = 2$ to $1.6\times$ at $\kappa = 40$. This is the expected behaviour of a first-order expansion rather than a defect, but it bounds the useful range, and a practitioner should read the correction as buying roughly an order of magnitude only while κ stays in the first half of this table.

Figure 2 shows the three layers together. Comparing layers A and B tests the discretization. The maximum discrepancy is 2.2×10^{-3} at $\kappa = 5$ and 1.6×10^{-3} at $\kappa = 20$, against a sampling noise of about 10^{-3} at 50,000 trajectories.

κ	softmax	with Green–Kubo	improvement
2	1.6×10^{-3}	5.0×10^{-5}	$31\times$
5	3.7×10^{-3}	3.0×10^{-4}	$12\times$
10	6.9×10^{-3}	1.1×10^{-3}	$6.3\times$
20	1.2×10^{-2}	3.9×10^{-3}	$3.2\times$
40	2.0×10^{-2}	1.2×10^{-2}	$1.6\times$

Table 1: Maximum absolute error in the win probabilities against the exactly discretized generator, over sixteen boundary windows. The correction gains an order of magnitude at small κ and degrades to nothing by $\kappa = 40$, which is where the neglected $O(\kappa^2)$ term takes over.

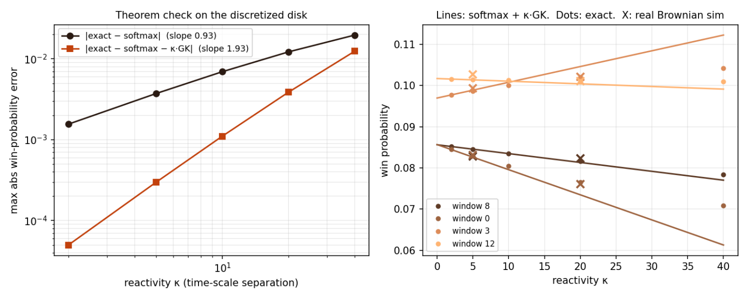

Figure 2: The theorem on continuous physics (experiment 9). Reflected Brownian motion in the unit disk with boundary reaction windows, error against κ for the leading and corrected laws, measured orders 0.93 and 1.93 against an exactly discretized Neumann generator.

Two systematic errors. Both would have corrupted a direct comparison of simulation against theory, and neither is visible without the middle layer.

The first is band quantization, worth 5×10^{-2}. Sharp cell indicators mis-size any window narrower than a grid cell, with the mis-sizing ratio running from 2.0 down to zero, since the narrowest window here spans about half a cell and is missed entirely. Fractional area-exact cell coverage fixes it, after which band areas match their analytic values to 2×10^{-17}.

The second is overlap semantics, worth 1.4×10^{-2}. The window geometry contains three overlapping pairs. The original simulation awarded an overlap to the lower-indexed window, while the model stacks intensities as $\lambda = \kappa \sum_i g_i$, which is the physically correct convention for independent receptors. The error's structure identified it, being -0.014 on one window and $+0.006$ on its overlap partner, and its flatness under both time-step and grid refinement confirmed that it was not a discretization artifact.

9 Reproducibility

Every number in this paper comes from a committed script. Experiment 7 gives the chain verification, experiment 9 the narrow-escape study, experiment 39 the runner-up bridge, experiment 40 the replication, the equal-time ablation, the rank bounds, and the index-order check, and experiment 41 the identifiability and estimation results. All four run from a clean checkout of `https://github.com/microprediction/kinetics` with NumPy, SciPy, and Matplotlib, and each writes the `results.csv` the corresponding numbers are quoted from. Experiments 7, 39, and 40 take seconds. Experiment 9 takes about three minutes. A regression suite locks every property stated here, including the two convergence orders, the exactness of common-mode cancellation, the rank bounds, and the runner-up identity.

10 Related work

The averaging that produces the leading term is classical (Khas'minskii, 1966; Kurtz, 1973; Pavliotis and Stuart, 2008; Yin and Zhang, 1998), and the closest recent instance is Colantoni (2026), who studies a killing rate modulated by a continuous-time Markov chain, gives Feynman–Kac representations, and proves an averaging principle in the fast-switching limit. That work stops where this one starts. It carries a single killing rate rather than competing channels, so there is no branching ratio to correct, and it establishes the limit without the first-order term.

The ingredient is classical on the other side too. Motional narrowing (Anderson, 1954; Kubo, 1954; van Kampen, 1974) and dynamical disorder (Zwanzig, 1990; Szabo et al., 1982) have used integrated rate autocovariances for seventy years, but for effective rates and survival probabilities, in the scalar setting where no choice law arises. Recent linear-response work on first-passage problems predicts the response of the mean first-passage *time* rather than the identity of the winner.

For marked Markov-modulated processes there are matrix-analytic methods (Fischer and Meier-Hellstern, 1993) that compute mark probabilities exactly, with no expansion and no asymptotic regime. Theorem 1 is the fast-environment asymptotic of those same quantities, and what it adds is not the numbers but the structure the asymptotic exposes. One pair $(\bar{\lambda}, K)$ serves every subset, shared modulation drops out, and low-rank environments give low-rank corrections.

That a race is at once a competing-risks model and a random-utility model is old. Marley and Colonius (1992) make the identification explicitly, recasting a Thurstonian race in cause-specific hazards, and prove the proportional-hazards equivalence that Proposition 2 recovers, building on the competing-risks results of Elandt-Johnson (1976) and Kochar and Proschan (1991). Their treatment of dependence runs the other way from ours. They record the classical non-identifiability, that any well-behaved structure on a fixed choice set admits an independent representation, so on a fixed set there is nothing to correct. The correction here lives in the comparison across sets, which is exactly where their context-free hazard condition bites. What appears not to have been stated is that softmax is the homogenized choice law of fast hidden kinetics, with a computable first correction.

11 Discussion

Two things come out of this. The expansion explains why softmax is a reasonable default in systems with fast shared dynamics and says how it fails, and the identification analysis says what an observer can hope to recover. The failure is governed by integrated dynamical correlations, not by static ones, and it is insensitive to any fluctuation shared across channels.

Four limitations bound the claim. The expansion is in the mixing-to-firing timescale ratio and says nothing when channels fire faster than the environment relaxes, which is the quenched limit of dynamical disorder (Zwanzig, 1990). The correction is first order, so systems at moderate separation retain an $O(\varepsilon^2)$ residual, quantified in Table 1, where the improvement decays to nothing by $\kappa = 40$. The theorem is proved for finite chains, and a formal treatment of reversible diffusions with a spectral gap is left open, so the narrow-escape study verifies rather than proves that case. The correction as usually quoted assumes the environment is observed in equilibrium, and Proposition 1 gives the extra term otherwise. Finally the continuum study is one geometry with one window arrangement, and while the chain results replicate over twenty environments, the physics results do not yet replicate over geometries.

One extension looks tractable. The disk's boundary operator diagonalizes in Fourier modes, which may render the leading geometric correction analytic rather than numerical.

Sections 5 and 6 close the gap between the computed correction and the estimated one. The identifiable part of K is characterized exactly, and ranked observations reach all of it. What remains is statistical rather than structural. The ε^{-2} sample requirement is severe at small ε, which is precisely the regime where the expansion is most accurate, so the theory is easiest to trust where it is hardest to fit. Whether the runner-up design also improves the constant in that rate, and not merely the rank, is open.

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